

Homework Set 3. *First-Order, Second-Order, and Higher-Order Systems* (Solutions)

Question 1

(a)

$$\theta = \begin{cases} \arctan(y/x), & x > 0 \\ \pi - \arctan(y/|x|), & x < 0, y > 0 \\ -\pi - \arctan(y/|x|), & x < 0, y \leq 0 \\ \frac{\pi}{2}, & x = 0, y > 0 \\ -\frac{\pi}{2}, & x = 0, y < 0 \end{cases}$$

(b)

i.

$$z_2 = \frac{1 - ai}{(1 + bi)(1 + ci)^2}$$

$$z_2 = r_2 e^{i\theta_2},$$

where

$$r_2 = |z_2| = \frac{|1 - ai|}{|1 + bi| \cdot |1 + ci|^2} = \frac{\sqrt{1 + a^2}}{\sqrt{1 + b^2} \cdot (1 + c^2)};$$

$$\theta_2 = \angle(z_2) = \angle(1 - ai) - \angle(1 + bi) - 2\angle(1 + ci) = -\arctan(a) - \arctan(b) - 2\arctan(c).$$

ii.

$$z_3 = \frac{(1 - bi)^3}{1 - a^2 + 2aci} \text{ (assuming } |a| < 1)$$

$$z_3 = r_3 e^{i\theta_3},$$

where

$$r_3 = |z_3| = \frac{|1 - bi|^3}{|1 - a^2 + 2aci|} = \frac{(1 + b^2)^{\frac{3}{2}}}{\sqrt{(1 - a^2)^2 + (2ac)^2}};$$

$$\theta_3 = \angle(z_3) = 3\angle(1 - bi) - \angle(1 - a^2 + 2aci) = 3(-\arctan(b)) - \arctan\left(\frac{2ac}{1 - a^2}\right).$$

iii.

$$z_4 = \frac{(1 - ai)(1 + bi)}{(1 + ai)^2} e^{b-ci}$$

$$z_4 = r_4 e^{i\theta_4}$$

where,

$$r_4 = |z_4| = \frac{|1 - ai| \cdot |1 + bi|}{|1 + ai|^2} \cdot |e^{b-ci}| = \frac{\sqrt{1 + a^2} \cdot \sqrt{1 + b^2}}{1 + a^2} \cdot e^b = \frac{e^b \sqrt{1 + b^2}}{\sqrt{1 + a^2}},$$

$$\begin{aligned} \theta_4 = \angle(z_4) &= \angle(1 - ai) + \angle(1 + bi) - 2\angle(1 + ai) + \angle(e^{b-ci}) \\ &= -\arctan(a) + \arctan(b) - 2\arctan(a) - c = \arctan(b) - 3\arctan(a) - c. \end{aligned}$$

Question 2

(a).

Calculate K. Since the denominator only has negative real roots, T(t) converges, so assume at t=30 min, the system converges to a constant temperature:

$$K = \frac{T_1(30) - T_1(0)}{\Delta p} = \frac{500 - 100 \text{ (}^\circ\text{C)}}{2 - 1 \text{ (kW)}} = 400 \left(\frac{^\circ\text{C}}{\text{kW}} \right)$$

The step input of power:

$$p'(t) = M \cdot S(t)$$

$$P'(s) = \frac{M}{s}$$

Where $M = 2 - 1 = 1 \text{ (kW)}$

Solve for $T'(t)$,

$$\frac{T'(s)}{P'(s)} = \frac{K}{\tau s + 1}$$

$$T'(s) = \frac{M}{s} \cdot \frac{K}{\tau s + 1} = KM \left(\frac{1}{s} - \frac{1}{s + \frac{1}{\tau}} \right)$$

$$T'(t) = KM \left(1 - e^{-\frac{t}{\tau}} \right)$$

Plug in $M = 1$, $K = 400$, and $T(t = 4 \text{ min}) = 400 \text{ (}^\circ\text{C)}$ to solve for τ . (T denotes original value, T' denotes deviation value, so $T'(t = 4 \text{ min}) = 400 - 100 = 300 \text{ (}^\circ\text{C)}$)

$$300 = 400 \left(1 - e^{-\frac{4}{\tau}} \right)$$

$$e^{-\frac{4}{\tau}} = \frac{1}{4}$$

$$-\frac{4}{\tau} = \ln \left(\frac{1}{4} \right)$$

$$\tau = 2.885 \text{ (min)}$$

(b)

The power input is changed to a ramp input:

$$p(t) = a \cdot t$$
$$P'(s) = \frac{a}{s^2}$$

Where $a = 0.5 \text{ kW/min}$.

$$T'(s) = \frac{a}{s^2} \cdot \frac{K}{\tau s + 1}$$

And use convolution formula to solve for $T'(t)$,

$$T'(t) = \int_0^t \frac{K}{\tau} \exp\left(-\frac{t-t'}{\tau}\right) \cdot at' dt'$$

$$T'(t) = aK \exp\left(-\frac{t}{\tau}\right) \cdot \int_0^t \exp\left(\frac{t'}{\tau}\right) \cdot t' d\left(\frac{t'}{\tau}\right)$$

$$T'(t) = aK \exp\left(-\frac{t}{\tau}\right) \cdot \int_0^t t' d\left(\exp\left(\frac{t'}{\tau}\right)\right)$$

$$T'(t) = aK \exp\left(-\frac{t}{\tau}\right) \cdot \left[t \exp\left(\frac{t}{\tau}\right) - 0 - \int_0^t \exp\left(\frac{t'}{\tau}\right) dt' \right]$$

$$T'(t) = aK \exp\left(-\frac{t}{\tau}\right) \cdot \left[t \exp\left(\frac{t}{\tau}\right) - \tau \int_0^t \exp\left(\frac{t'}{\tau}\right) d\left(\frac{t'}{\tau}\right) \right]$$

$$T'(t) = aK \exp\left(-\frac{t}{\tau}\right) \cdot \left[t \exp\left(\frac{t}{\tau}\right) - \tau \exp\left(\frac{t}{\tau}\right) + \tau \right]$$

$$T'(t) = aK \cdot \left[t - \tau + \tau \cdot \exp\left(-\frac{t}{\tau}\right) \right]$$

$$T'(t) = 200 \left(\frac{^{\circ}\text{C}}{\text{min}}\right) \cdot \left[t - 2.885 + 2.885 \cdot \exp\left(-\frac{t}{2.885(\text{min})}\right) \right] (\text{min})$$

To find the changing rate of temperature, we take derivative of the function:

$$\frac{dT'}{dt}(t) = 200 \left(\frac{^{\circ}\text{C}}{\text{min}}\right) \cdot \left[1 - \exp\left(-\frac{t}{2.885}\right) \right]$$

We can say the rate of change approaches its maximum as the exponential term approaches zero, i.e., as t becomes large and approaches ∞ .

Question 3

(a)

For a unit step input of height M , then $U(s) = M/s$. Thus,

$$Y(s) = G(s)U(s) = \frac{10M}{s(5s+1)(3s+1)}$$

Since the limit exists for all $a = \text{Re}(s) \geq 0$ in $\lim_{s \rightarrow a} (s Y(s))$, we can use the Final Value Theorem (FVT):

$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} (s Y(s))$; thus,

$$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} \left(s \frac{10M}{s(5s+1)(3s+1)} \right)$$
$$= \lim_{s \rightarrow 0} \left(\frac{10M}{(5s+1)(3s+1)} \right)$$

$$= \boxed{10M}$$

(b)

For a unit impulse, then $U(s) = 1$. Thus,

$$Y(s) = G(s)U(s) = \frac{10M}{(5s+1)(3s+1)}$$

Using the Final Value Theorem,

$$\begin{aligned} \lim_{t \rightarrow \infty} y(t) &= \lim_{s \rightarrow 0} \left(s \frac{10M}{(5s+1)(3s+1)} \right) \\ &= \boxed{0} \end{aligned}$$

Question 4

(a)

$$\text{Gain} = \frac{11.2\text{mm} - 8\text{mm}}{3\text{ lpsi} - 15\text{psi}} = 0.20\text{mm/psi}$$

$$\text{Overshoot} = \frac{12.7\text{mm} - 11.2\text{mm}}{11.2\text{mm} - 8\text{mm}} = 0.47$$

$$\text{Overshoot} = \exp\left(\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}\right) = 0.47, \quad \zeta = 0.234$$

$$\text{Period} = \left(\frac{2\pi\tau}{\sqrt{1-\zeta^2}}\right) = 2.3 \text{ sec}$$

$$\tau = 2.3 \text{ sec} \times \frac{\sqrt{1-0.234^2}}{2\pi} = 0.356 \text{ sec}$$

$$\frac{R'(s)}{P'(s)} = \frac{0.2}{0.127s^2 + 0.167s + 1} \quad (1)$$

(b)

From Eq. 1, taking the inverse Laplace transform,

$$0.127 \ddot{R}' + 0.167 \dot{R}' + R' = 0.2 P'$$

$$\ddot{R}' = \ddot{R} \quad \dot{R}' = \dot{R} \quad R' = R - 8 \quad P' = P - 15$$

$$0.127 \ddot{R} + 0.167 \dot{R} + R = 0.2 P + 5$$

$$\ddot{R} + 1.31 \dot{R} + 7.88 R = 1.57 P + 39.5$$

Question 5

(a)

$$p'(t) = (6-4) S(t), \quad P'(s) = \frac{2}{s}$$

$$Q'(s) = \frac{-3}{20s+1} P'(s) = \frac{-3}{20s+1} \frac{2}{s}$$

$$q'(t) = -6(1 - e^{-t/20})$$

(b)

$$R'(s) + Q'(s) = P'_m(s)$$

$$r'(t) + q'(t) = p'_m(t) = p_m(t) - p_m(0)$$

$$r'(t) = p_m(t) - 12 + 6(1 - e^{-t/20})$$

$$K = \frac{r'(t=\infty)}{p(t=\infty) - p(t=0)} = \frac{18-12+6(1-0)}{6-4} = 6$$

Overshoot,

$$OS = \frac{r'(t=15) - r'(t=\infty)}{r'(t=\infty)} = \frac{27-12+6(1-e^{-15/20})-12}{12} = 0.514$$

$$OS = \exp\left(\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}\right) = 0.514, \quad \zeta = 0.2$$

Period T for $r'(t)$ is equal to the period for $p_m(t)$ because $e^{-t/20}$ decreases monotonically.

$$\text{Thus, } T = 50 - 15 = 35$$

$$\text{and } \tau = \frac{T}{2\pi} \sqrt{1-\zeta^2} = 5.46$$

(c)

$$\frac{P'_m(s)}{P'(s)} = \frac{K}{\tau^2 s^2 + 2\zeta\tau s + 1} + \frac{K'}{\tau' s + 1}$$

$$= \frac{(K'\tau^2)s^2 + (K\tau' + 2K'\zeta\tau)s + (K + K')}{(\tau^2 s^2 + 2\zeta\tau s + 1)(\tau' s + 1)}$$

(d)

$$\text{Overall process gain} = \left. \frac{P'_m(s)}{P'(s)} \right|_{s=0} = K + K' = 6 - 3 = 3 \frac{\%}{\text{psi}}$$

Question 6

(a)

For this process with two CSTRs in series and a recycle loop, we have a single reaction: $A \xrightarrow{k} B$ with reaction rate: $r = kc_A$. For species A, $r_A = \nu_A r = -kc_A$. Note that c_0 , c_1 , and c_2 are the concentrations of species A at different locations in the process. F_0 , F_1 , and F_2 are constant volumetric flowrates. The flowrate of the recycle stream is represented by the product of the recycle ratio and the effluent stream volumetric flowrate, RF_2 , and is constant.

Model for Tank 1

$$\begin{aligned} & \{\text{accumulation of species A in Tank 1}\} \\ &= \{\text{A flowing into Tank 1}\} - \{\text{A flowing out of Tank 1}\} \\ &+ \{\text{A gen./cons. by rxn in Tank 1}\} \end{aligned}$$

Let the subscript A denote the composition of the variable corresponding to species A.

$$\begin{aligned} \frac{dN_A}{dt} &= \dot{n}_{A,\text{inlet stream}} + \dot{n}_{A,\text{recycle stream}} - \dot{n}_{A,\text{out}} + Vr_A \\ \frac{dN_A}{dt} &= F_0c_0 + RF_2c_2 - F_1c_1 - Vr_A \end{aligned}$$

Since $N_A = Vc_A$ and V is constant

$$\frac{dc_A}{dt} = \frac{F_0c_0 + RF_2c_2 - F_1c_1}{V} - r_A = \frac{F_0}{V}c_0 + \frac{RF_2}{V}c_2 - \frac{F_1}{V}c_1 - kc_A$$

For a well-mixed CSTR, the concentration at the outlet is equal to the tank concentration. $c_A = c_1$. Thus,

$$\frac{dc_1}{dt} = \frac{F_0}{V}c_0 + \frac{RF_2}{V}c_2 - \frac{F_1}{V}c_1 - kc_1$$

Since $F_1 = RF_2 + F_2$ and $F_0 = F_2$, then $F_1 = F_0(R + 1)$

$$\begin{aligned} \frac{dc_1}{dt} &= \frac{F_0}{V}c_0 + \frac{RF_0}{V}c_2 - \frac{F_0(R + 1)}{V}c_1 - kc_1 \\ \frac{dc_1}{dt} &= \frac{F_0}{V}c_0 - \left(\frac{F_0}{V}(R + 1) + k\right)c_1 + \frac{RF_0}{V}c_2 \end{aligned}$$

Model for Tank 2

$$\begin{aligned} \frac{dN_A}{dt} &= \dot{n}_{A,\text{in}} - \dot{n}_{A,\text{out}} + Vr_A \\ \frac{dN_A}{dt} &= F_1c_1 - (F_2c_2 + RF_2c_2) + Vr_A \\ \frac{dN_A}{dt} &= F_1c_1 - F_2(R + 1)c_2 + Vr_A \end{aligned}$$

$N_A = Vc_A$, V is constant, and for a well-mixed CSTR, the concentration at the outlet is equal to the tank concentration. $c_A = c_2$. Thus,

$$\begin{aligned}\frac{dc_2}{dt} &= \frac{F_1}{V} c_1 - \frac{F_2}{V} (R + 1) c_2 - k c_2 \\ \frac{dc_2}{dt} &= \frac{F_1}{V} c_1 - \left(\frac{F_2}{V} (R + 1) + k \right) c_2\end{aligned}$$

Since $F_0 = F_2$, then $F_1 = F_0(R + 1)$,

$$\frac{dc_2}{dt} = \frac{F_0}{V} (R + 1) c_1 - \left(\frac{F_0}{V} (R + 1) + k \right) c_2$$

Models for Both Tanks

Two simultaneous equations:

$$\begin{aligned} * \quad \frac{dc_1}{dt} &= \frac{F_0}{V} c_0 - \left(\frac{F_0}{V} (R + 1) + k \right) c_1 + \frac{RF_0}{V} c_2 \\ * \quad \frac{dc_2}{dt} &= \frac{F_0}{V} (R + 1) c_1 - \left(\frac{F_0}{V} (R + 1) + k \right) c_2\end{aligned}$$

Where F_0 , R , V , and k are constants. c_0 is the process input, and c_1 and c_2 are the dependent variables.

Steady-State Condition:

Setting the time derivatives to zero gives the steady-state condition

$$\begin{aligned} * \quad 0 &= \frac{F_0}{V} \bar{c}_0 - \left(\frac{F_0}{V} (R + 1) + k \right) \bar{c}_1 + \frac{RF_0}{V} \bar{c}_2 \\ * \quad 0 &= \frac{F_0}{V} (R + 1) \bar{c}_1 - \left(\frac{F_0}{V} (R + 1) + k \right) \bar{c}_2\end{aligned}$$

Deviation Variable Form

Let $c'_i = c_i - \bar{c}_i$ represent the deviation variable for concentration where $\bar{c}_i$ is the steady-state value of said concentration, then, the two equations become the following:

$$\begin{aligned}\frac{d(c'_1 + \bar{c}_1)}{dt} &= \frac{F_0}{V} (c'_0 + \bar{c}_0) - \left(\frac{F_0}{V} (R + 1) + k \right) (c'_1 + \bar{c}_1) + \frac{RF_0}{V} (c'_2 + \bar{c}_2) \\ \frac{d(c'_2 + \bar{c}_2)}{dt} &= \frac{F_0}{V} (R + 1) (c'_1 + \bar{c}_1) - \left(\frac{F_0}{V} (R + 1) + k \right) (c'_2 + \bar{c}_2)\end{aligned}$$

Simplifying the constant out of the derivatives and combining steady-state terms gives the following:

$$\begin{aligned}\frac{d(c'_1)}{dt} &= \frac{F_0}{V} c'_0 - \left(\frac{F_0}{V} (R + 1) + k \right) c'_1 + \frac{RF_0}{V} c'_2 + \left\{ \frac{F_0}{V} \bar{c}_0 - \left(\frac{F_0}{V} (R + 1) + k \right) \bar{c}_1 + \frac{RF_0}{V} \bar{c}_2 \right\} \\ \frac{d(c'_2)}{dt} &= \frac{F_0}{V} (R + 1) c'_1 - \left(\frac{F_0}{V} (R + 1) + k \right) c'_2 + \left\{ \frac{F_0}{V} (R + 1) \bar{c}_1 - \left(\frac{F_0}{V} (R + 1) + k \right) \bar{c}_2 \right\}\end{aligned}$$

Note that the terms in the curly braces are zero, since that is the defined steady-state condition. This leaves the following two simultaneous equations in deviation-variable form

$$\boxed{\frac{d(c'_1)}{dt} = \frac{F_0}{V} c'_0 - \left(\frac{F_0}{V} (R + 1) + k \right) c'_1 + \frac{RF_0}{V} c'_2}$$

$$\frac{d(c_2')}{dt} = \frac{F_0}{V} (R + 1) c_1' - \left(\frac{F_0}{V} (R + 1) + k \right) c_2'$$

(b)

Note that the differential equations are linear; thus, they can be readily manipulated into transfer functions. Let $C_i(s) = \mathcal{L}[c_i']$, then the Laplace transforms of both equations are as follows (assuming that the process is initially at steady state):

$$sC_1(s) = \frac{F_0}{V} C_0(s) - \left(\frac{F_0}{V} (R + 1) + k \right) C_1(s) + \frac{RF_0}{V} C_2(s)$$

$$sC_2(s) = \frac{F_0}{V} (R + 1) C_1(s) - \left(\frac{F_0}{V} (R + 1) + k \right) C_2(s)$$

Via algebraic manipulation of the two equations above, we can find the transfer function relating the outlet concentration to the inlet concentration: $\frac{C_2(s)}{C_0(s)}$.

$$\frac{C_2(s)}{C_0(s)} = \frac{F_0^2 (R + 1)}{V^2 s^2 + 2(F_0(R + 1) + Vk)Vs + F_0^2 (R + 1) + 2F_0(R + 1)Vk + V^2 k^2}$$

(c)

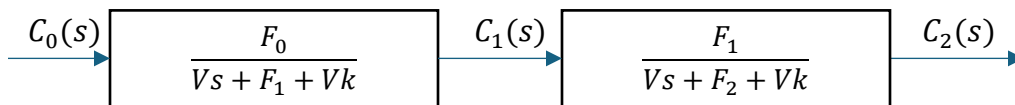
In the limit as $R \rightarrow 0$, then,

$$\begin{aligned} \frac{C_2(s)}{C_0(s)} &= \frac{F_0^2}{V^2 s^2 + 2(F_0 + Vk)Vs + F_0^2 + 2F_0Vk + V^2 k^2} \\ &= \frac{F_0^2}{V^2 s^2 + 2(F_0 + Vk)Vs + (F_0 + Vk)^2} \\ &= \frac{F_0^2}{(Vs + F_0 + Vk)^2} \\ &= \left(\frac{F_0}{Vs + F_0 + Vk} \right) \left(\frac{F_0}{Vs + F_0 + Vk} \right) \end{aligned}$$

We note that as $R \rightarrow 0$, $F_1 = F_0(R + 1) \rightarrow F_0$, and remember that $F_0 = F_2$ then

$$\frac{C_2(s)}{C_0(s)} = \left(\frac{F_0}{Vs + F_1 + Vk} \right) \left(\frac{F_1}{Vs + F_2 + Vk} \right)$$

The above boxed equation represents is transfer function product is identical to two CSTRs in series with no recycle as shown below:



(d)

If no reaction is occurring ($k = 0$), and at infinite recycle ($R \rightarrow \infty$), then,

$$\frac{C_2(s)}{C_0(s)} = \lim_{R \rightarrow \infty} \left\{ \frac{F_0^2(R+1)}{V^2 s^2 + 2F_0(R+1)Vs + F_0^2(R+1)} \right\}$$

Using L'Hôpital's rule and taking the partial derivative w.r.t. R in the numerator and denominator:

$$\begin{aligned} &= \lim_{R \rightarrow \infty} \left\{ \frac{F_0^2}{2F_0Vs + F_0^2} \right\} \\ &= \frac{F_0}{2Vs + F_0} \end{aligned}$$

Remembering that $F_0 = F_2$,

$$\boxed{\frac{C_2(s)}{C_0(s)} = \frac{F_0}{2Vs + F_2}}$$

Note this is identical to a transfer function of a single tank with inlet volumetric flowrate, F_0 , outlet volumetric flowrate, F_2 , and volume, $2V$.

To show that the gain is equivalent to one, we rearrange the transfer function such that it is in the form:

$\frac{K}{\tau s + 1}$ where K is the gain.

$$\frac{C_2(s)}{C_0(s)} = \frac{F_0/F_2}{\frac{2V}{F_2}s + 1}$$

Here we see that the gain is $K = \frac{F_0}{F_2}$. Since $F_0 = F_2$, then, $\boxed{K = 1}$

Question 7

(a) The form of the transfer function of Tank 1 and Tank 2:

$$\frac{T_1'(s)}{Q'(s)} = \frac{K_1}{\tau_1 s + 1} \quad \frac{T_2'(s)}{T_1'(s)} = \frac{K_2}{\tau_2 s + 1}$$

Since the two tanks are in series, the overall transfer function:

$$\frac{T_1'(s)}{Q'(s)} \cdot \frac{T_2'(s)}{T_1'(s)} = \frac{K_1 K_2}{(\tau_1 s + 1)(\tau_2 s + 1)}$$

By an inspection of the data, $\tau_1 > \tau_2$. Following Eq. 6-58 in Seborg do the following approximation:

$$\frac{T_1'(s)}{Q'(s)} \cdot \frac{T_2'(s)}{T_1'(s)} \approx \frac{K_1 K_2 e^{-\tau_2 s}}{\tau_1 s + 1}$$

Calculate static gain K_1 and K_2 , then use the data to fit Eq. 7-17:

$$\begin{aligned} K_1 &= \frac{T_1(50) - T_1(0)}{\Delta q} = \frac{18 - 10}{85 - 82} = 2.667 \\ K_2 &= \frac{T_2(50) - T_2(0)}{T_1(50) - T_1(0)} = \frac{26 - 20}{18 - 10} = 0.75 \end{aligned}$$

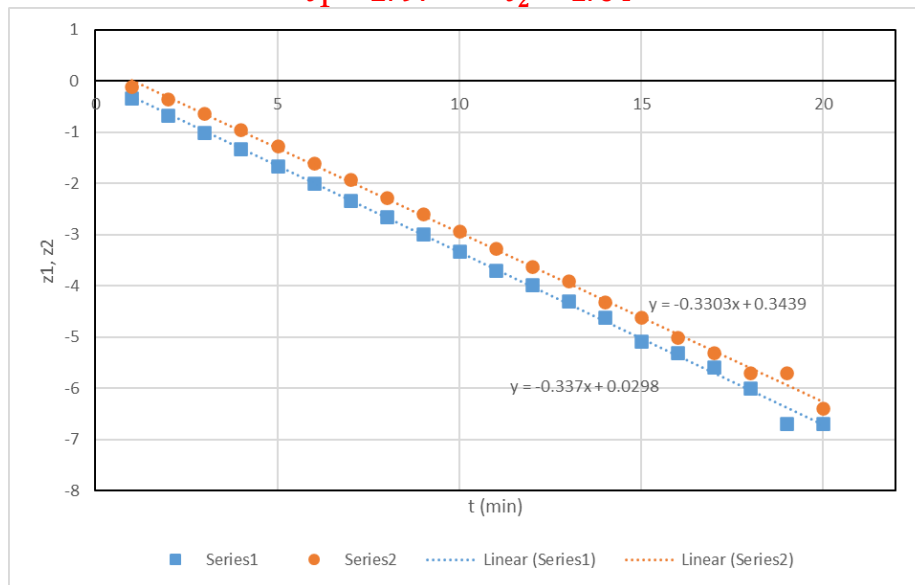
$$z_1(t) = \ln \left[\frac{T_1(50) - T_1(t)}{T_1(50) - T_1(0)} \right] = -\frac{t - \theta}{\tau_1} = -0.337 \cdot t + 0.0298$$

$$z_2(t) = \ln \left[\frac{T_2(50) - T_2(t)}{T_2(50) - T_2(0)} \right] = -\frac{t - \tau_2}{\tau_1} = -0.3303 \cdot t + 0.3439$$

$$-\frac{1}{\tau_1} = -0.337$$

$$0.3303 \cdot \tau_2 = 0.3439$$

$$\tau_1 = 2.97 \quad \tau_2 = 1.04$$



$$\frac{T'_1(s)}{Q'(s)} = \frac{2.667}{2.97s + 1}$$

$$\frac{T'_2(s)}{T'_1(s)} = \frac{0.75}{1.04s + 1}$$

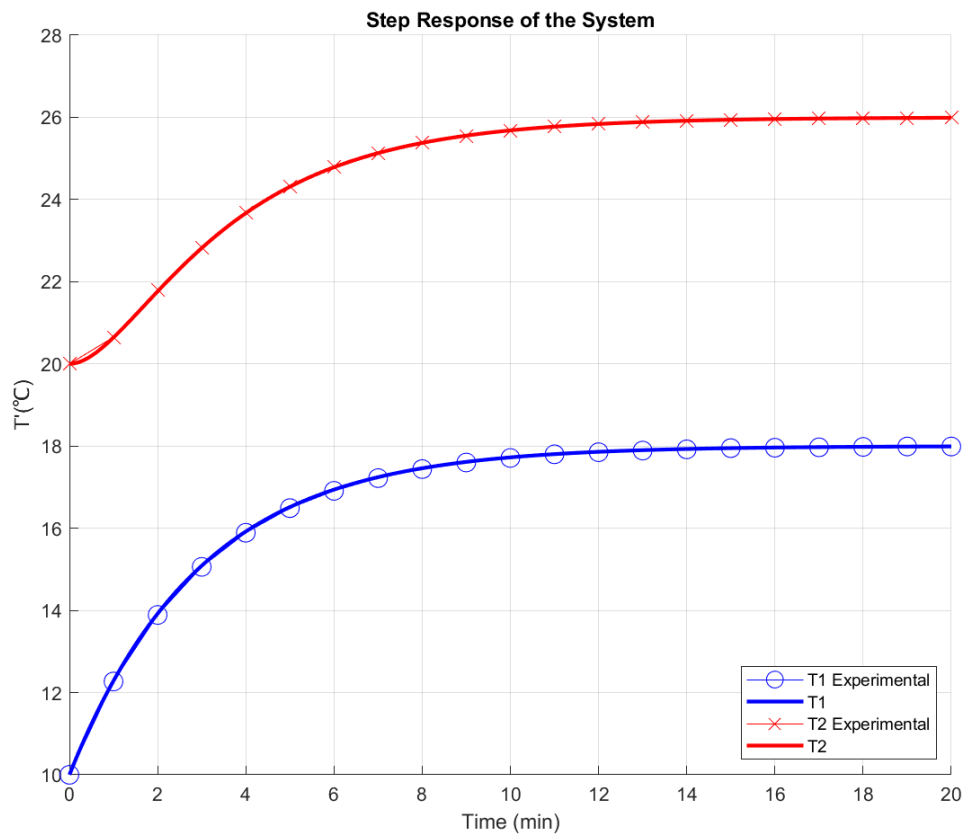
(b) The MATLAB code for generating the plots is provided below

```
% Experiments data
t_data = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20]; %
Time in minutes
T1_data = [10, 12.27, 13.89, 15.06, 15.89, 16.49, 16.91, 17.22, 17.44, 17.6, 17.71,
17.8, 17.85, 17.89, 17.92, 17.95, 17.96, 17.97, 17.98, 17.99, 17.99];
T2_data
=[20, 20.65, 21.79, 22.83, 23.68, 24.32, 24.79, 25.13, 25.38, 25.55, 25.68, 25.77, 25.84, 25.88, 25
.92, 25.94, 25.96, 25.97, 25.98, 25.98, 25.99];
% solve T1 transfer function %
sys1 = tf(2.667, [2.97, 1]); % Transfer Function of Tank 1
t1 = 0:0.1:20; % Time vector from 0 to 20 sec
[y1, t1] = step(3*sys1, t1); % Step response with magnitude 3
% solve T2 transfer function
```

```

sys2 = tf(0.75, [1.04, 1]); % Transfer Function of Tank 2
sys3 = series(sys1, sys2); % Transfer Functions in series
t2 = 0:0.1:20; % Time vector from 0 to 20 sec
[y2, t2] = step(3*sys3, t2); % Step response with magnitude 3
% Plot
figure;
hold on;
plot(t_data, T1_data, 'bo-', 'MarkerSize', 10, 'DisplayName', 'T1 Experimental');
plot(t1, y1 + 10, 'b', 'LineWidth', 2, 'DisplayName', 'T1');
plot(t_data, T2_data, 'rx-', 'MarkerSize', 10, 'DisplayName', 'T2 Experimental');
plot(t2, y2 + 20, 'r', 'LineWidth', 2, 'DisplayName', 'T2');
xlabel('Time (min)');
ylabel('T(°C)');
ylim([10, 28]);
title('Step Response of the System');
legend('Location', 'southeast');
grid on;
hold off;

```



Question 8

$$\frac{Y(s)}{X(s)} = \frac{K(\tau_a s + 1)}{(\tau_1 s + 1)(\tau_2 s + 1)}, \quad \tau_1 > \tau_2, \quad X(s) = M/s$$

$$y(t) = KM \left[1 + \frac{\tau_a - \tau_1}{\tau_1 - \tau_2} e^{-t/\tau_1} - \frac{\tau_a - \tau_2}{\tau_1 - \tau_2} e^{-t/\tau_2} \right]$$

(a)

$$\text{Extremum} \Rightarrow \dot{y}(t) = 0$$

$$KM \left[0 - \frac{1}{\tau_1} \left(\frac{\tau_a - \tau_1}{\tau_1 - \tau_2} \right) e^{-t/\tau_1} + \frac{1}{\tau_2} \left(\frac{\tau_a - \tau_2}{\tau_1 - \tau_2} \right) e^{-t/\tau_2} \right] = 0$$

$$\frac{1 - \tau_a / \tau_2}{1 - \tau_a / \tau_1} = e^{-t \left(\frac{1}{\tau_1} - \frac{1}{\tau_2} \right)} \geq 1 \quad \text{since} \quad \tau_1 > \tau_2$$

(b)

$$KM \left[1 + \frac{\tau_a - \tau_1}{\tau_1 - \tau_2} e^{-t/\tau_1} - \frac{\tau_a - \tau_2}{\tau_1 - \tau_2} e^{-t/\tau_2} \right] > KM$$

$$\frac{\tau_a - \tau_1}{\tau_a - \tau_2} > e^{-t \left(\frac{1}{\tau_2} - \frac{1}{\tau_1} \right)} > 0, \quad \text{therefore} \quad \tau_a > \tau_1$$

(c)

$$\text{Inverse response} \Rightarrow \dot{y}(t) < 0 \quad \text{at} \quad t = 0^+$$

$$KM \left[0 - \frac{1}{\tau_1} \left(\frac{\tau_a - \tau_1}{\tau_1 - \tau_2} \right) e^{-t/\tau_1} + \frac{1}{\tau_2} \left(\frac{\tau_a - \tau_2}{\tau_1 - \tau_2} \right) e^{-t/\tau_2} \right] < 0 \quad \text{at} \quad t = 0^+$$

$$- \frac{1}{\tau_1} \left(\frac{\tau_a - \tau_1}{\tau_1 - \tau_2} \right) + \frac{1}{\tau_2} \left(\frac{\tau_a - \tau_2}{\tau_1 - \tau_2} \right) < 0$$

$$\frac{\tau_a \left(\frac{1}{\tau_2} - \frac{1}{\tau_1} \right)}{\tau_1 - \tau_2} < 0$$

$$\text{Since } \tau_1 > \tau_2, \quad \tau_a < 0.$$

(d)

If an extremum in y exists, then from (a):

$$e^{-t\left(\frac{1}{\tau_1}-\frac{1}{\tau_2}\right)}=\left(\frac{1-\tau_a/\tau_2}{1-\tau_a/\tau_1}\right)$$

$$t=\frac{\tau_1\tau_2}{\tau_1-\tau_2}\ln\left(\frac{1-\tau_a/\tau_2}{1-\tau_a/\tau_1}\right)$$